

Introduction

Outline

- Why is Human-Computer Interaction Important?
- Terminology - Basics
- Terminology - UX
- User Needs



Why is Human-Computer Interaction Important?

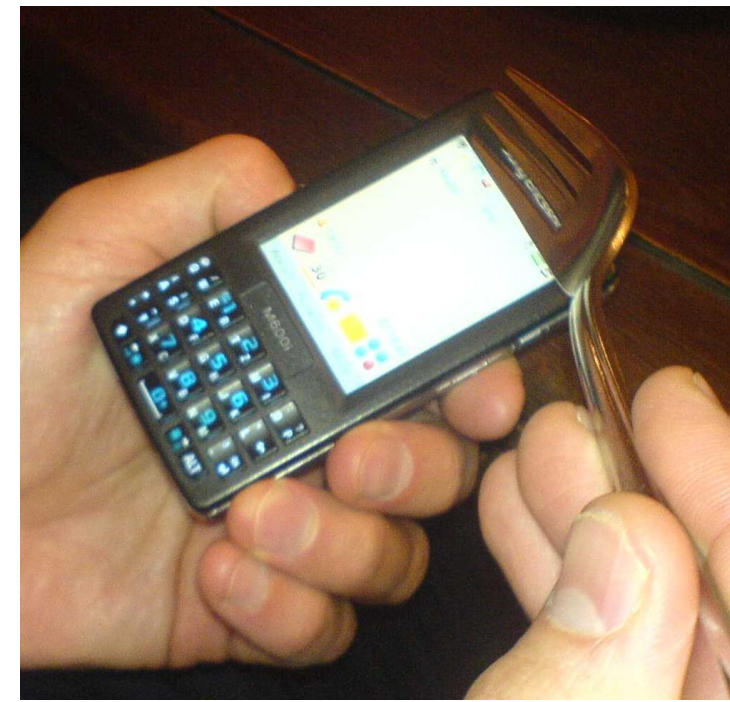
Learning Goals

- Understand ...
 - how user interfaces impacts our interaction and relationship with (digital) technologies
 - why interaction is relevant from different perspectives (user, companies)
 - why designing interaction is becoming more relevant as technology advances
- Be able to ...
 - explain the economic implications of bad usability
 - sketch a user interface and discuss different aspects
 - explain why interaction is a visible innovation

Human-Computer Interaction

How does Human-Computer Interaction impact us?

- It determines how we use (digital) products.
- For examples it impacts...
 - what we can do with products and services,
 - how easy or hard it is to work with a software,
 - how quickly you can learn to use a system, or
 - how safe a product is.
- It is central to how we feel and what we experience while interacting with digital technologies.



Why is Usability Important?

Products that are easy to use a good for business.

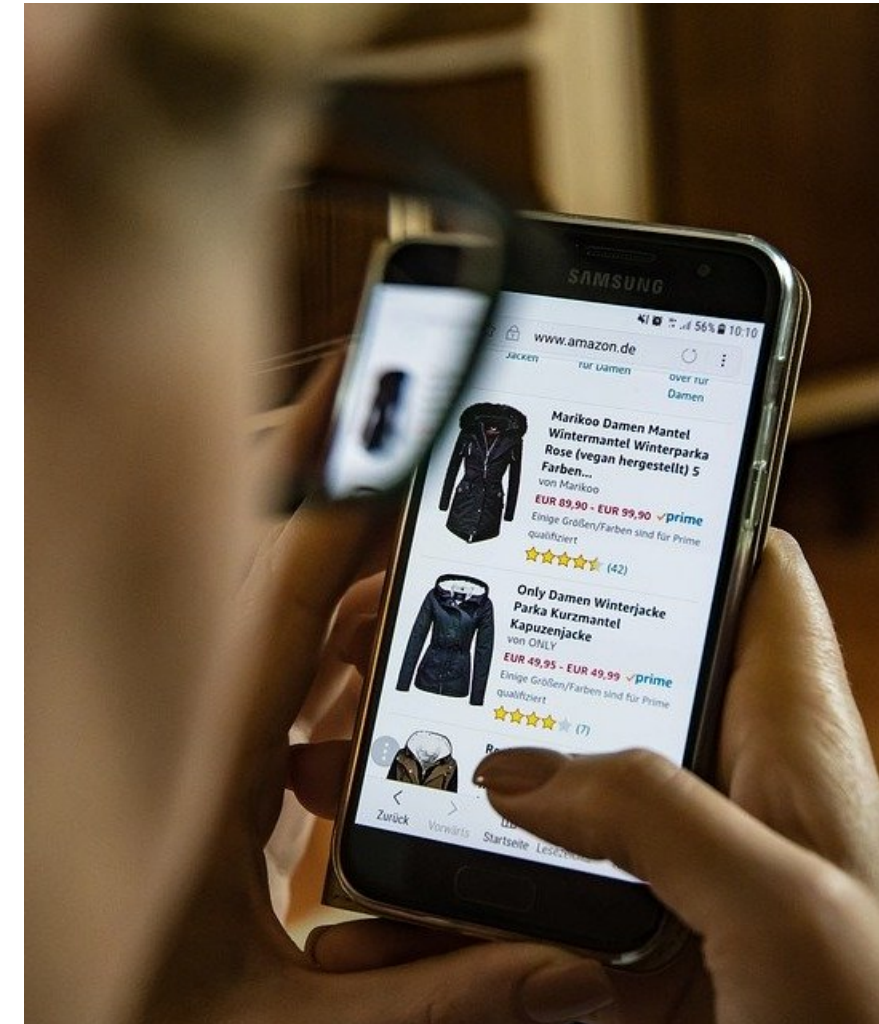
- Improving usability can
 - increase productivity of users,
 - reduce costs (support, efficiency),
 - increase sales/revenue (web shop),
 - enhance customer loyalty, or
 - win new customers.
- Case studies that show the benefit of usability
- Usability is often considered as sign of quality
- Usability gives a competitive advantage.



Why Focus on User and Interaction?

How to discriminate your product or service?

- Traditionally product discrimination is by functionality and price
- Why should a customer pick you, if competitors offer very similar functions (e.g. messaging services) at similar prices (e.g. payed by advertising)?
 - Your product is easier to use?
 - You get your tasks done faster?
 - You have more fun doing what you want to do?



Why Focus on User and Interaction?

Trends that make Human-Computer interaction more important

- It becomes harder to discriminate by technology
- Availability of bandwidth, storage and processing
- New input and output technologies
- Computing becomes part of many traditional devices
- Willingness for training or learning applications decreases
- Life-style technologies are more and more digital
- Broad and diverse user groups
- Majority of user are not interested in the technology

- New understanding of computing
 - The old question: What can computers do?
 - The new question: What can humans do with computers?
 - Book by Ben Shneiderman: Leonardo's Laptop

B. Shneiderman. Leonardo's Laptop: Human Needs and the New Computing Technologies. 2002.

Economic dimension of Usability?

Importance in the Context of WWW, Apps and New Media?

- User Interface is often the central discriminating factor
- Often the same product/service is sold at very similar prices
- Competition is very close (just another App, browser tab, ...)
- Comparison is easily possible
- Examples: Online-Shop
 - Users who cannot find the product in the shop cannot buy it
 - Users who cannot fill in the payment form are not going to buy
 - Users who worry if the item fits them are less likely to buy
 - Typically a direct correlation between usability and sales
- “Bad Usability is Like a Leaky Pipe”
<https://90percentofeverything.com/2006/11/13/bad-usability-is-like-a-leaky-pipe/>



Concerns in Human-Computer Interaction

Science, Engineering, and Design Aspects

- The joint performance of tasks by humans and machines
- The structure of communication between humans and machines
- Human capabilities to use machines (including the learnability of interfaces)
- Engineering concerns that arise in designing and building interfaces
- The process of specification, design, and implementation of interfaces
- Design trade-offs
- Algorithms and programming of the user interface itself

Example: Sketch an Interface

How would a user interface that includes both functions look like?
Make sketches for 2 alternative interfaces.

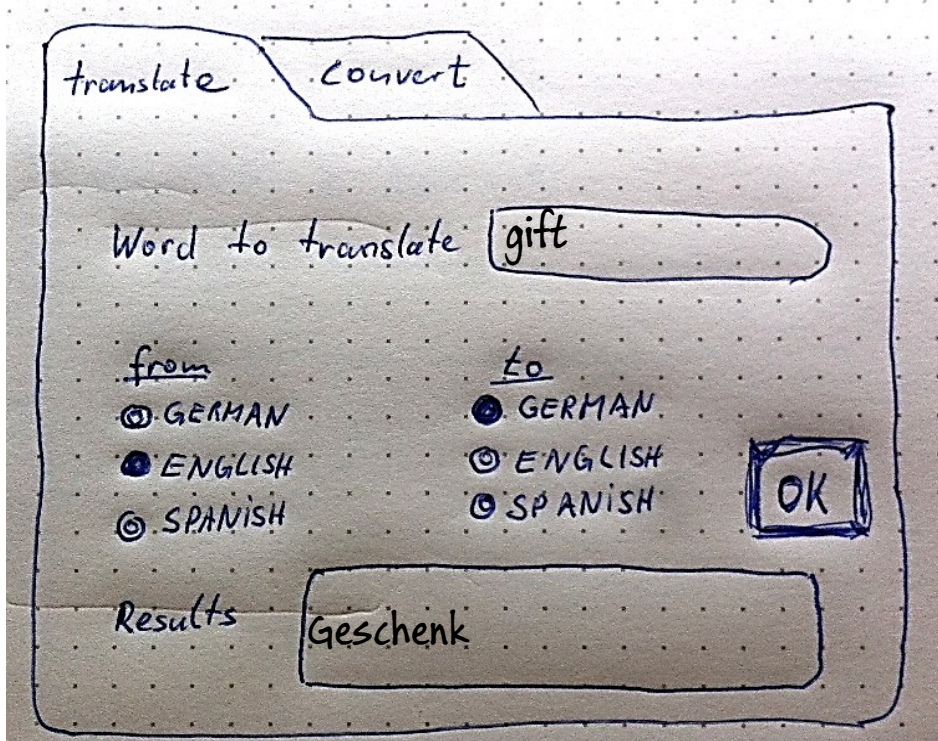
```
String translate(int fromLanguage, int toLanguage, String str)
// translate is a function that takes a word (str)
// the word is from a specific language (fromLanguage) and it
// translates it into a word into another language (toLanguage)
// Example: wordInSpanish = convert(1, 3, "Haus")
// fromLanguage is the language the word is from, toLanguage
// ist die target language (1=German, 2=English, 3=Spanish)

float convert(int fromCurrency, int toCurrency, float amount)
// convert is a function that takes a number (amount) of a specific
// currency (fromCurrency) and it converts it into a number
// (toCurrency) representing the amount in the target currency
// Example: myDollar = convert(6, 7, 19.23)
// fromCurrency ist initlal currency and toCurrency is the target
// currency (6=Euro, 7=US Dollar, 8=Britisches Pfund)
```

Example: Sketch an Interface

How would a user interface that includes both functions look like?
Make sketches for 2 alternative interfaces.

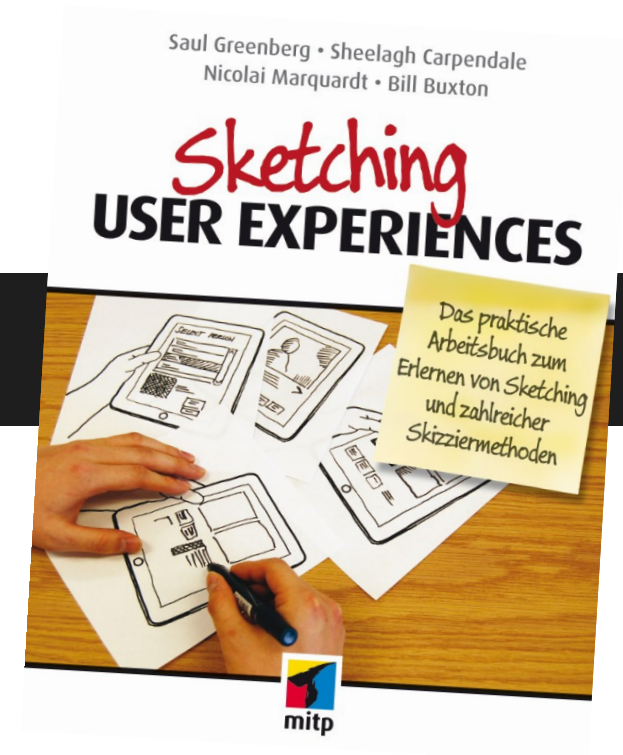
```
String translate(int fromLanguage, int toLanguage, String str)
float convert(int fromCurrency, int toCurrency, float amount)
```



Example: Sketch an Interface

How would a user interface that includes both functions look like?
Make sketches for 2 alternative interfaces.

```
String translate(int fromLanguage, int toLanguage, String str)
float convert(int fromCurrency, int toCurrency, float amount)
```



<https://www.mitp.de/IT-WEB/Software-Entwicklung/Sketching-User-Experiences.html>

Example: Sketch an Interface

How would a user interface that includes both functions look like?
Make sketches for 2 alternative interfaces.

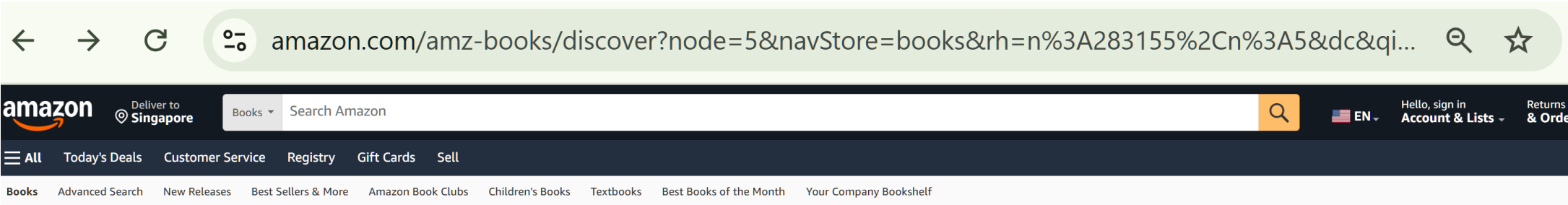
```
String translate(int fromLanguage, int toLanguage, String str)
float convert(int fromCurrency, int toCurrency, float amount)
```



<https://www.goodreads.com/book/show/8014393-sketching-user-experiences>

It is NOT Only About the User Interface!

Experience includes the overall product and service



New releases

[See more new releases >](#)

Superagency: What Could Possibly Go Right with...	The Technological Republic: Hard Power,...	Some Future Day: How AI Is Going to Change...	Beyond Cracking the Coding Interview: Pass...	Superbloom: How Technologies of...	HBR Guide to Generative AI for...	Genesis: Artificial Intelligence, Hope, and th...	Windows 11 for Seniors: Complete Step-by-St
4.7★★★★☆ 44		4.4★★★★☆ 36	4.5★★★★☆ 21	4.7★★★★☆ 13	5.0★★★★☆ 3	4.2★★★★☆ 362	5.0★★★★☆ 10
Hardcover	Hardcover	Hardcover	Paperback	Hardcover	Paperback	Hardcover	Paperback
-26% \$23.71 List Price: \$32.00	\$27.00	\$29.69	\$42.70	\$24.59	\$20.85	\$27.00	\$26.85

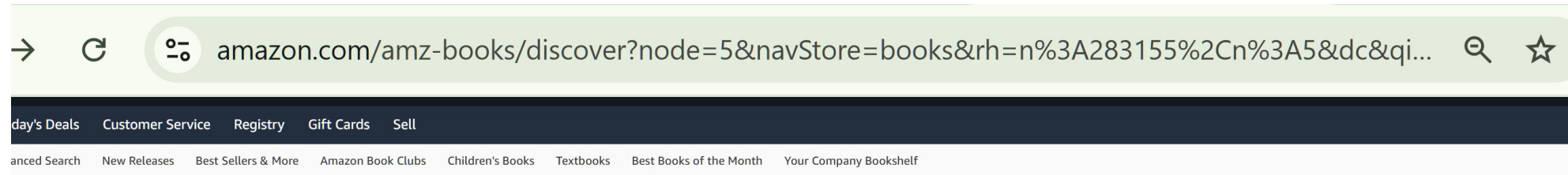
Popular

[See more popular >](#)

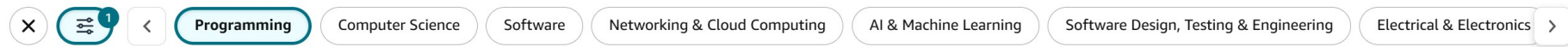


It is NOT Only About the User Interface!

Experience includes the overall product and service

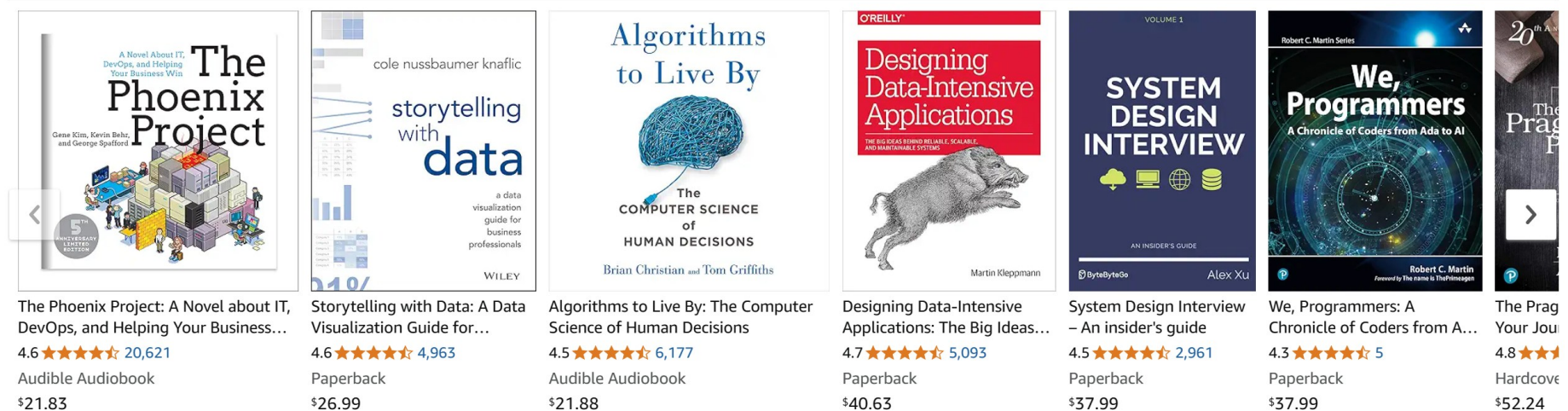


Explore Computers & Technology



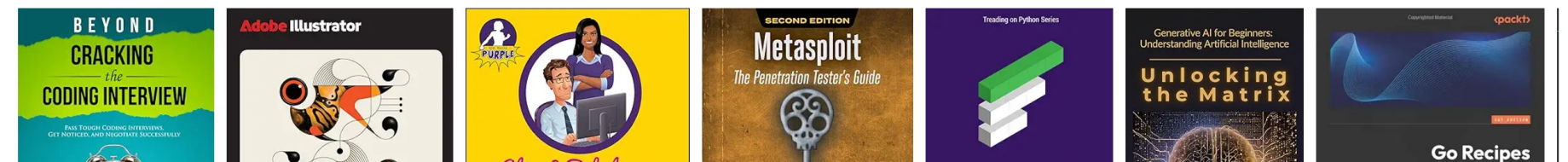
Popular

[See more popular >](#)



New releases

[See more new releases >](#)



Why is Hu

Human-Computer Interaction

...it is about visible innovation

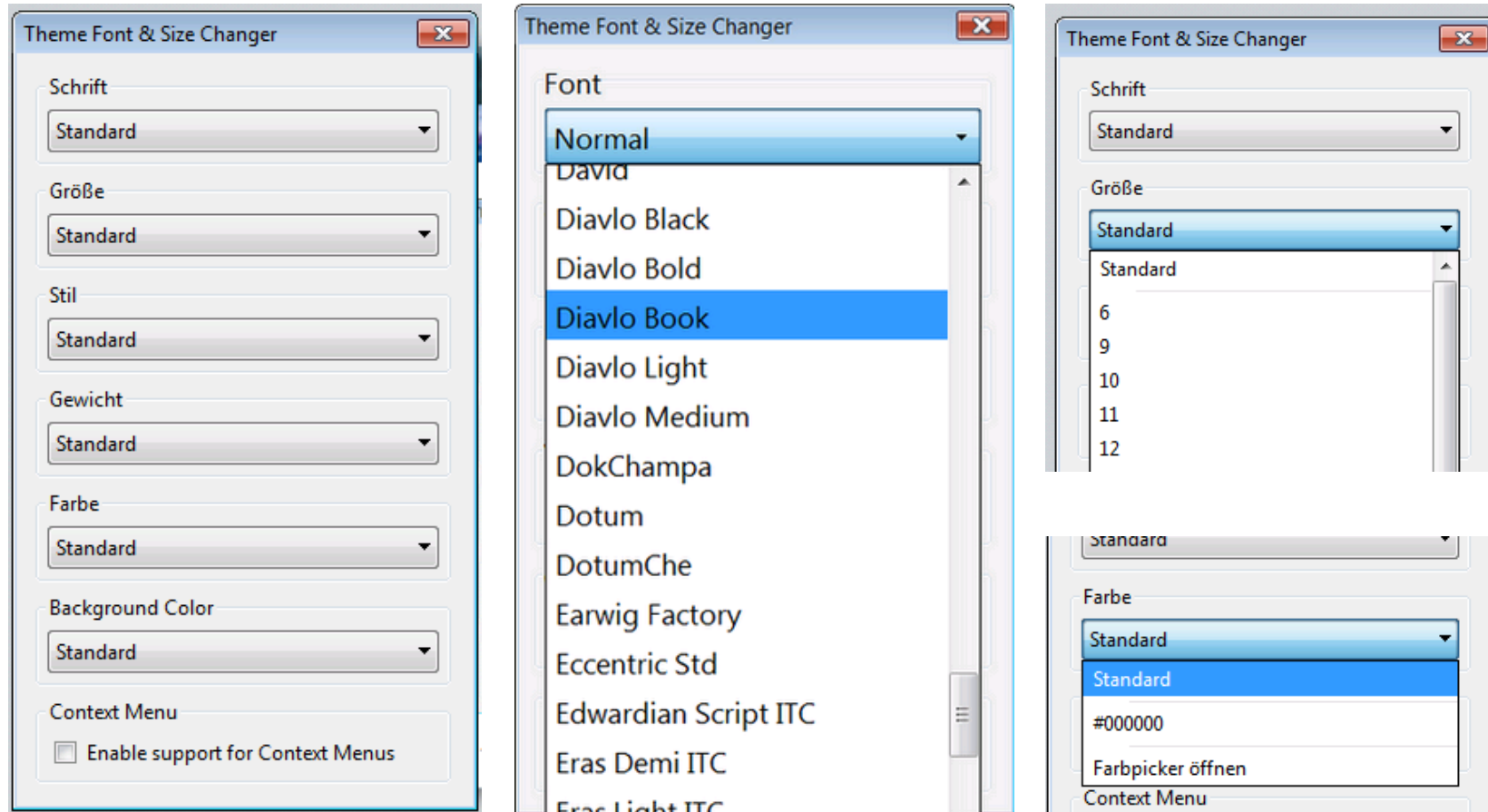
What is your Solution?

- Problems and challenges in Human-Computer Interaction
 - often not seen as problem before there is a solution
 - if problems are identified, they are typically easy to understand by non-experts
- Solutions in Human-Computer Interaction
 - once a solution is there, people will generally not remember that there was a problem
 - good and in particular great solutions (if found) often appear obvious (once they are there)
- The step from problem to solution is however not trivial (but this is often forgotten, once there is a solution)



Example: Selection/Menu for Fonts

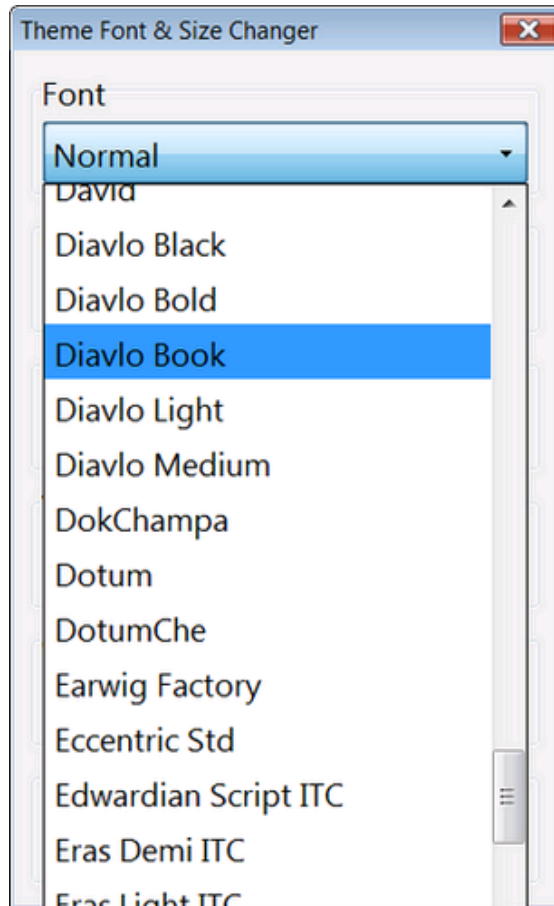
Is there a Problem? How to improve it?



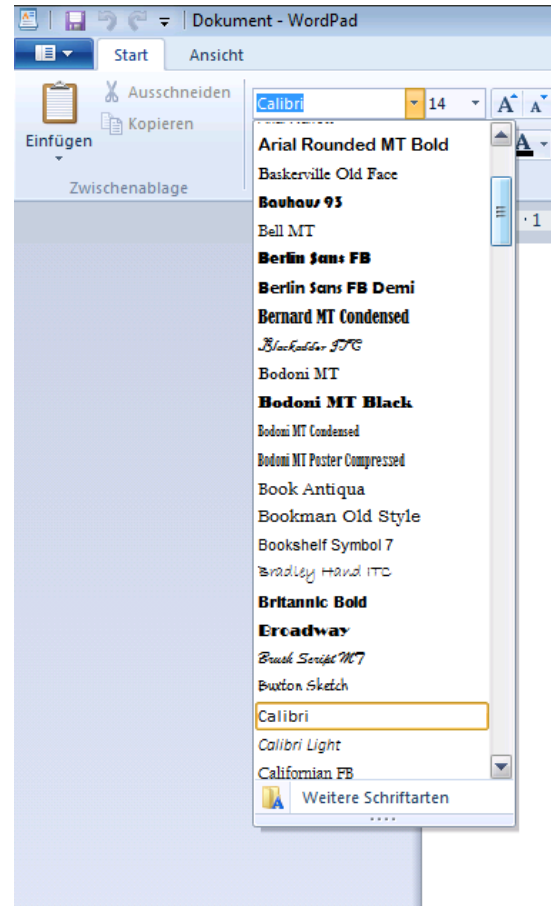
<https://addons.mozilla.org/en-US/firefox/addon/theme-font-size-changer/>

Example: Selection/Menu for Fonts

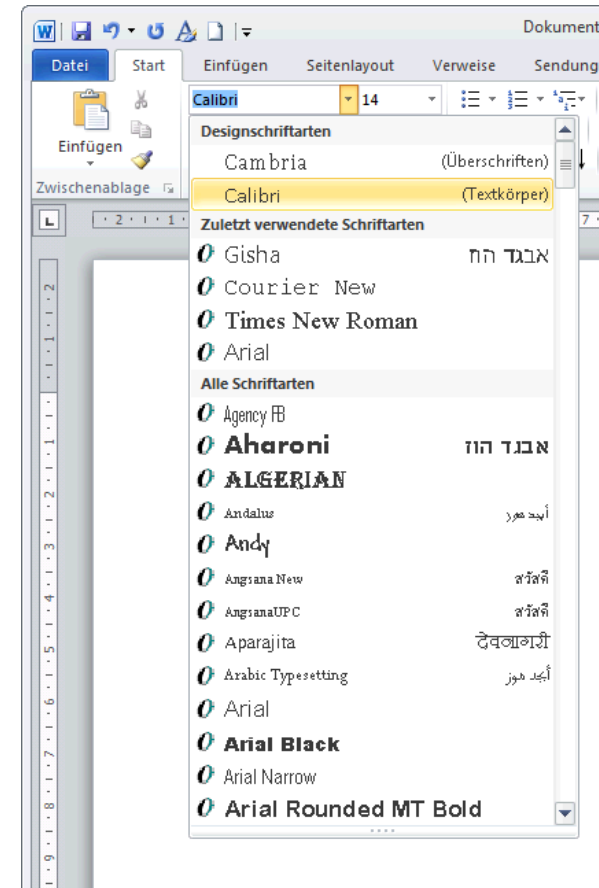
Is there a Problem? How to improve it?



<https://addons.mozilla.org/en-US/firefox/addon/theme-font-size-changer/>



WordPad/Win7

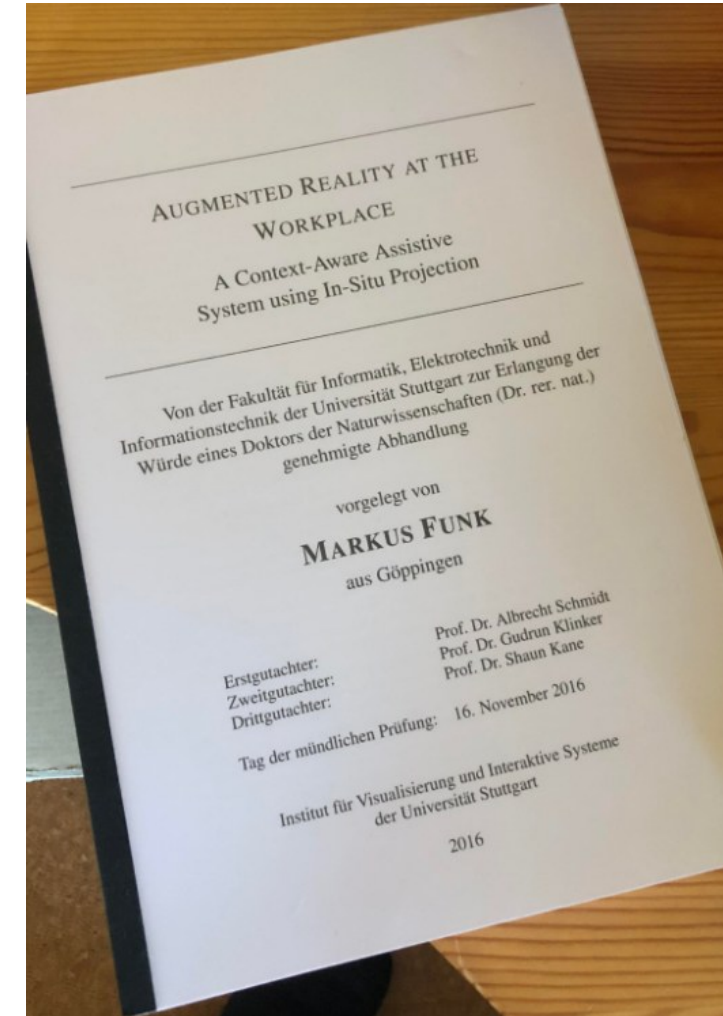


Microsoft Office Professional Plus 2010

Discussion: How long does it take to write a Thesis or a Book?

On a PC vs. on a Phone

- Assume the following
 - the book has 40.000 words
 - typing speed for physical keyboard is 40 words per minute
 - typing speed for phone keyboard is 15 words per minute
- Does this calculation make sense?
- What are we missing here?
- What is different when we write a text message?



Summary

The interface between the user and the information has become most critical for creating effective, efficient and pleasant systems.

- Interface and interaction design defines how we can use devices, services, and applications.
- The interface and interaction design impacts the performance of the user.
- It is important to understand how people use a system as a tool to achieve their goals.
- Human-computer interaction is relevant as it becomes harder to differentiate products based on features.
- Good usability is economically important.
- Sketching user interfaces and discussing design options and trade-offs is important.
- Innovation in Human-Computer interaction is mostly visible.



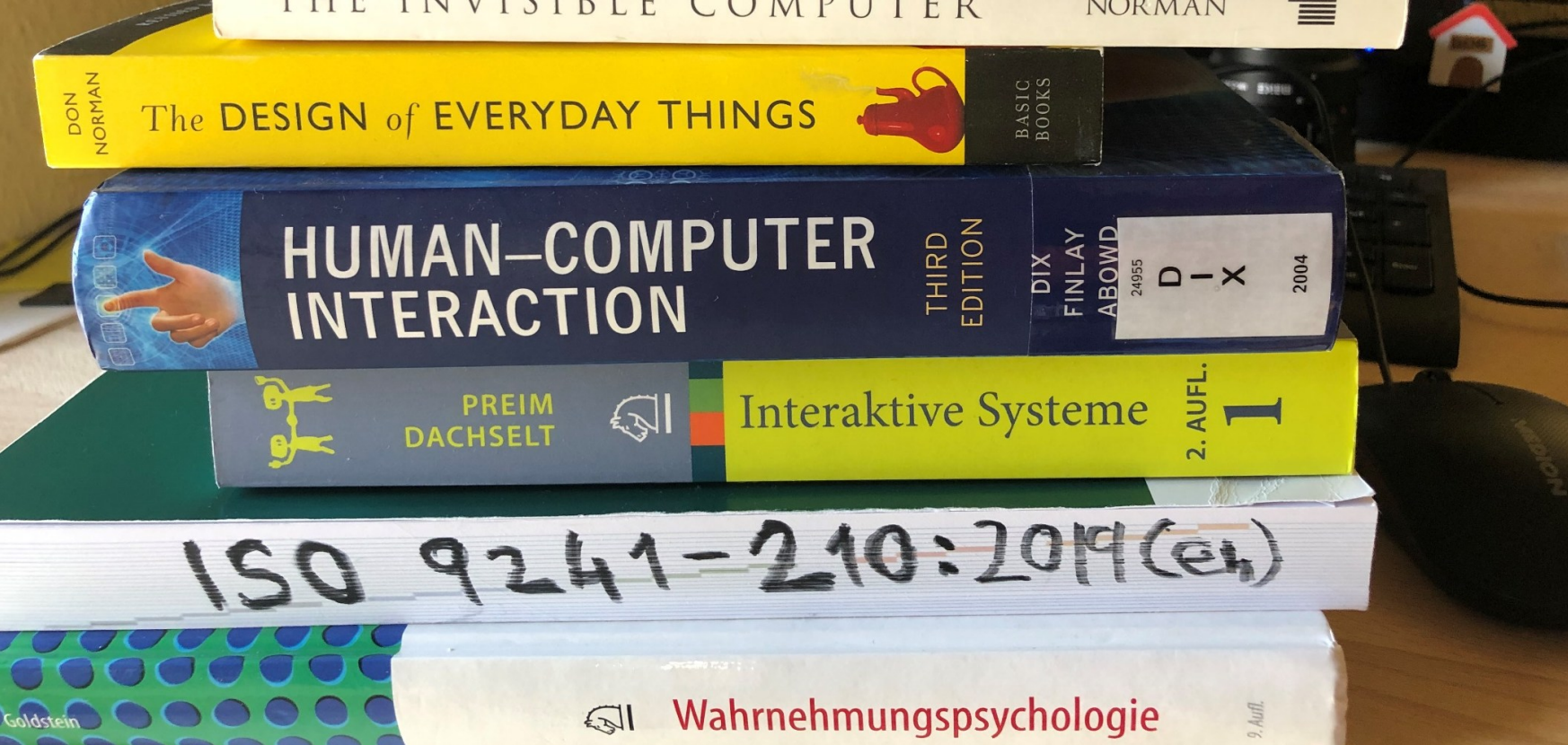
Did you understand this lesson?

Can you answer these questions?

- How does the user interface and interaction design impact the user? Give specific examples.
- Explain the economic benefits of a product that is easy to use in the context of a Web based shoe shop.
- What advantages can you expect if you improve the usability of an App or web page?
- Name some trends that make Human-Computer interaction more important.
- Sketch 2 alternative (and different) user interfaces for a world time clock app for a smart watch and discuss the advantages and disadvantages of the designs
- Give an example that highlights the statement “Problems in HCI are often not seen as problem before there is a solution.”

References

- B. Shneiderman (2002). Leonardo's Laptop: Human Needs and the New Computing Technologies.
<http://mitpress.mit.edu/main/feature/leonardoslaptop/index.html>
- B. Buxton (2010). *Sketching user experiences: getting the design right and the right design*. Morgan Kaufmann.
- Harry Brignull (2006). Bad usability is like a leaky pipe.
<https://90percentofeverything.com/2006/11/13/bad-usability-is-like-a-leaky-pipe/>
- Credits: Prof. Albrecht Schmidt



Noun [edit]

interaction (*countable and uncountable, plural interactions*)

1. The **situation** or **occurrence** in which two or more **objects** or **events** act upon one another to produce a new **effect**; the effect resulting from such a situation or occurrence. [quotations ▼]

*Be aware of **interactions** between different medications.*

2. A **conversation** or **exchange** between people.

*I enjoyed the **interaction** with a bunch of like-minded people.*

Terminology - Basics

Learning Goals

- Understand ...
 - technical terms that are central to human computer interaction,.
 - descriptions and texts that use these terms,
 - what user interface design is.
- Be able to explain technical terms, e.g.
 - Human-Computer Interaction,
 - Interactive Systems,
 - User Interface, or
 - Utility, Usability, Likeability.
- Know definitions relevant to HCI

Human-Computer Interaction

Definition

- “Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them”
(definition in the ACM SIGCHI Curricula for HCI, 1992)
- A pragmatic computer science view point:
How to enable interaction between one or more humans and one or more computational devices and with data.

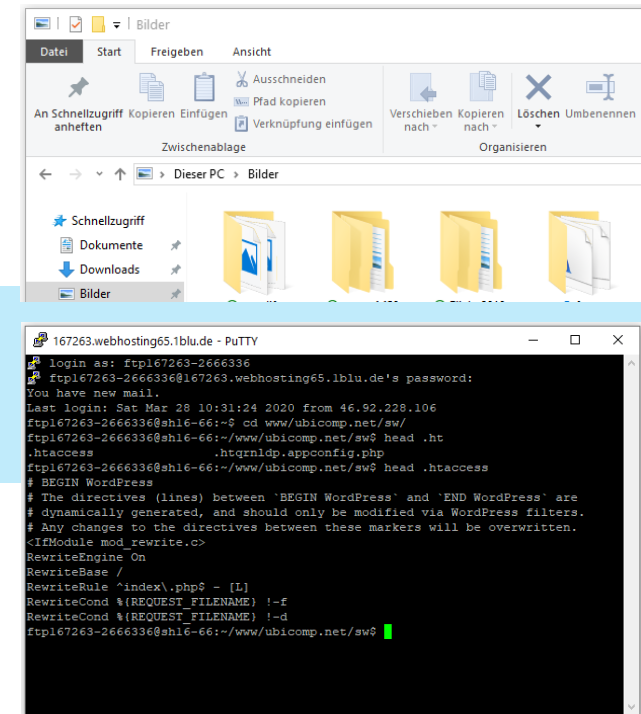
Thomas T. Hewett, Ronald Baecker, Stuart Card, Tom Carey, Jean Gasen, Marilyn Mantei, Gary Perlman, Gary Strong, and William Verplank. 1992. ACM SIGCHI Curricula for Human-Computer Interaction. Technical Report. Association for Computing Machinery, New York, NY, USA.

Interactive Systems

Definition and Examples

- An Interactive System is a computational system that allows users to interact in real-time¹. Interactions receive instant feedback visible to the user.
- Interactive computing is used in a similar way, with a focus that is less on the systems aspect.
- Examples
 - Graphical user interfaces, such as Windows 10 or MacOS
 - Mobile devices, such as an Android phone
 - gaming consoles, such as Xbox with Kinect or Nintendo Switch
 - Ticket vending machines, such as the DB-ticket machine
 - Command line interfaces, such as an SSH console

¹ The term 'real-time' refers here to system responses that are in a time range where users perceive a instant answer. Practically this is typically under 100ms, see <https://www.nngroup.com/articles/powers-of-10-time-scales-in-ux/>

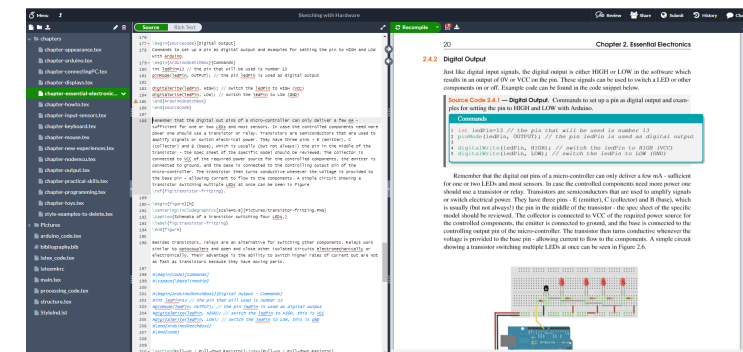
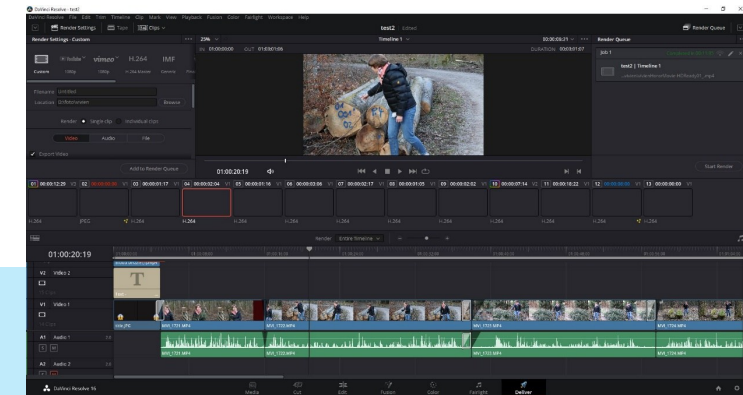


What is an Interactive Systems?

Mini-Exercise

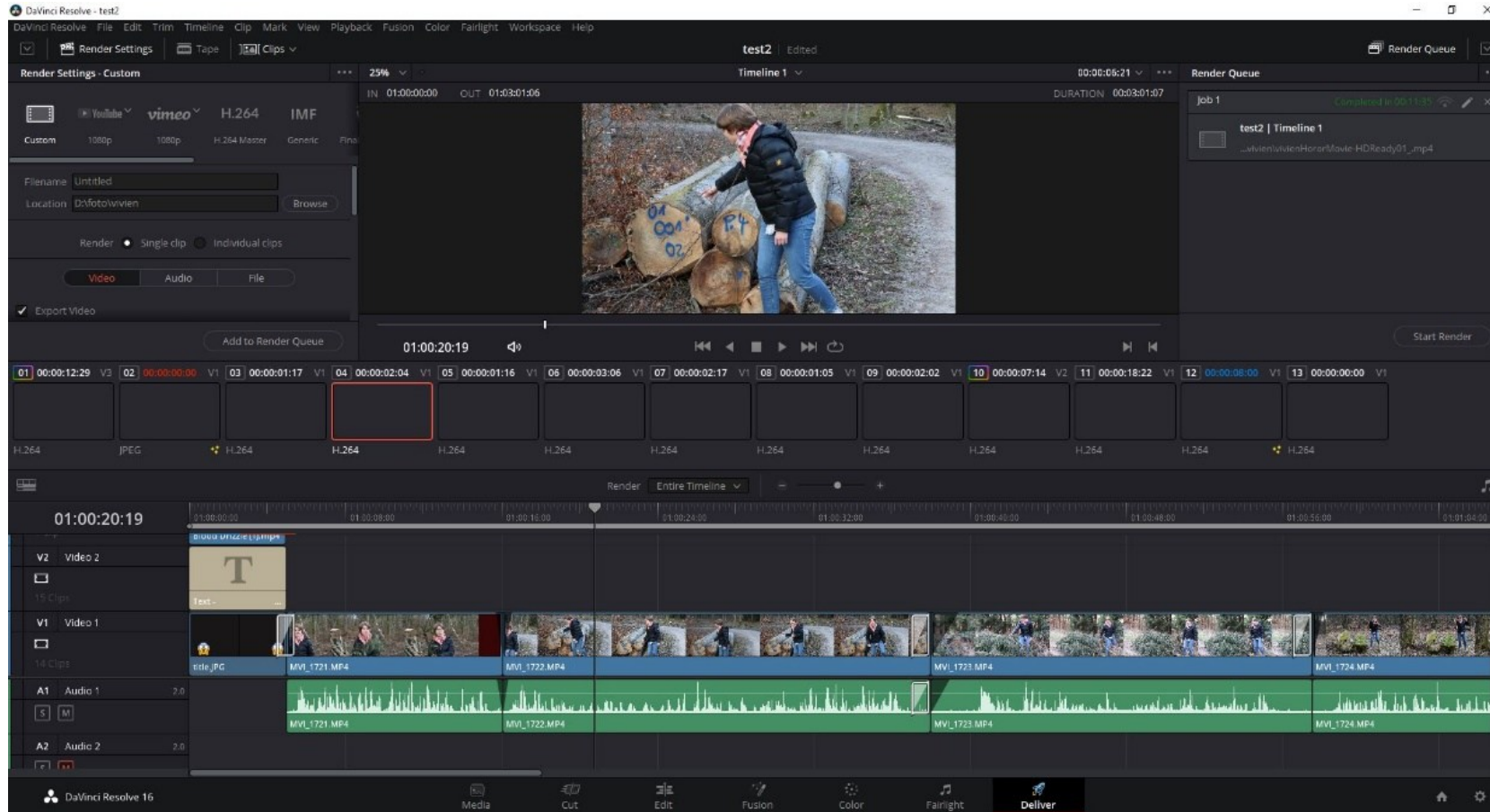
- Which of the following devices and applications can be considered an interactive System? *
 - Why can they be considered an interactive systems?
 - Why may they be not considered an interactive system?
- Discuss one of the following
 - Professional video editing tool
 - Electronic signage at the railway station
 - Latex text processing system
 - 3D printer

* There may be not 'yes' or 'no' answer. Consider different aspects and think how the design impacts whether or not this is perceived as an interactive system.



What is an Interactive Systems?

Mini-Exercise



What is an Interactive Systems?

Mini-Exercise



What is an Interactive Systems?

Mini-Exercise

Menu

Sketching with Hardware

Review Share Submit History Chat

chapters

chapter-appearance.tex

chapter-arduino.tex

chapter-connectingPC.tex

chapter-displays.tex

chapter-essential-electronic...

chapter-howto.tex

chapter-input-sensors.tex

chapter-keyboard.tex

chapter-mouse.tex

chapter-new-experiences.tex

chapter-nodemcu.tex

chapter-output.tex

chapter-practical-skills.tex

chapter-programming.tex

chapter-toys.tex

style-examples-to-delete.tex

Pictures

arduino_code.tex

bibliography.bib

latex_code.tex

latexmkrc

main.tex

processing_code.tex

structure.tex

StyleInd.lst

Source Rich Text

176

177 - \begin{sourcecode}[Digital Output]

178 Commands to set up a pin as digital output and examples for setting the pin to HIGH and LOW with Arduino.

179 - \begin{ArduinoSketchBox}[Commands]

180 int ledPin=13 // the pin that will be used is number 13

181 pinMode(ledPin, OUTPUT); // the pin ledPin is used as digital output

182

183 digitalWrite(ledPin, HIGH); // switch the ledPin to HIGH (VCC)

184 digitalWrite(ledPin, LOW); // switch the ledPin to LOW (GND)

185 \end{ArduinoSketchBox}

186 \end{sourcecode}

187

188 Remember that the digital out pins of a micro-controller can only deliver a few mA - sufficient for one or two LEDs and most sensors. In case the controlled components need more power one should use a transistor or relay. Transistors are semiconductors that are used to amplify signals or switch electrical power. They have three pins - E (emitter), C (collector) and B (base), which is usually (but not always!) the pin in the middle of the transistor - the spec sheet of the specific model should be reviewed. The collector is connected to VCC of the required power source for the controlled components, the emitter is connected to ground, and the base is connected to the controlling output pin of the micro-controller. The transistor then turns conductive whenever the voltage is provided to the base pin - allowing current to flow to the components. A simple circuit showing a transistor switching multiple LEDs at once can be seen in Figure

189 \ref{fig:transistor-fritzing}.

190 - \begin{figure}[h]

191 \centering\includegraphics[scale=0.6]{Pictures/transistor-fritzing.PNG}

192 \caption{Schemata of a transistor switching four LEDs.}

193 \label{fig:transistor-fritzing}

194 \end{figure}

195

196 Besides transistors, relays are an alternative for switching other components. Relays work similar to optocouplers and open and close other isolated circuits electromechanically or electronically. Their advantage is the ability to switch higher rates of current but are not as fast as transistors because they have moving parts.

197

198 \begin{code}[Commands]

199 \vspace{\baselineskip}

200

201 \begin{ArduinoSketchBox}[Digital Output - Commands]

202 \int ledPin=13 // the pin that will be used is number 13

203 \pinMode(ledPin, OUTPUT); // the pin ledPin is used as digital output

204 \digitalWrite(ledPin, HIGH); // switch the ledPin to HIGH, this is VCC

205 \digitalWrite(ledPin, LOW); // switch the ledPin to LOW, this is GND

206 \end{ArduinoSketchBox}

207 \end{code}

208

209

210 - \section{Pull-up / Pull-down Resistor}\index{Pull-up / Pull-down Resistor}

20

Chapter 2. Essential Electronics

2.4.2 Digital Output

Just like digital input signals, the digital output is either HIGH or LOW in the software which results in an output of 0V or VCC on the pin. These signals can be used to switch a LED or other components on or off. Example code can be found in the code snippet below.

Source Code 2.4.1 — Digital Output. Commands to set up a pin as digital output and examples for setting the pin to HIGH and LOW with Arduino.

Commands

1 int ledPin=13 // the pin that will be used is number 13

2 pinMode(ledPin, OUTPUT); // the pin ledPin is used as digital output

3

4 digitalWrite(ledPin, HIGH); // switch the ledPin to HIGH (VCC)

5 digitalWrite(ledPin, LOW); // switch the ledPin to LOW (GND)

Remember that the digital out pins of a micro-controller can only deliver a few mA - sufficient for one or two LEDs and most sensors. In case the controlled components need more power one should use a transistor or relay. Transistors are semiconductors that are used to amplify signals or switch electrical power. They have three pins - E (emitter), C (collector) and B (base), which is usually (but not always!) the pin in the middle of the transistor - the spec sheet of the specific model should be reviewed. The collector is connected to VCC of the required power source for the controlled components, the emitter is connected to ground, and the base is connected to the controlling output pin of the micro-controller. The transistor then turns conductive whenever the voltage is provided to the base pin - allowing current to flow to the components. A simple circuit showing a transistor switching multiple LEDs at once can be seen in Figure 2.6.

User Interface

Definition

- The part of a system where a user can interact with a system, device or application.
- The user interfaces can support input, output or both.
- User interfaces are not restricted to digital technologies or interactive systems
- Similar terms:
 - Human-computer interface (HCI)
 - Man-machine interface (MMI) or Human-Machine-Interface (HMI)
- Examples:
 - Graphical user interface (GUI) on a web page
 - Voice user interface in a smart speaker (e.g. Alexa)
 - Buttons, switches, wheels and levers in a cockpit



User, Interactive System, and User Interface

Definitions according to ISO 9241-210:2019(en)

- User: “person who interacts with a system, product or service”
- Context of Use: “ combination of users, goals and tasks, resources, and environment”
- Interactive System: “combination of hardware and/or software and/or services and/or people that users interact with in order to achieve specific goals”
- User Interface: “all components of an interactive system (software or hardware) that provide information and controls for the user to accomplish specific tasks with the interactive system”

<https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:v1:en>

User Interface Design and Interaction Design

Definition

- **User Interface Design** refers to the conception and design of the **user interface**, including the choice of **modalities**, selection of **interface elements** and their **placement** in the interface.
- **Interaction Design:**
 - Preece, Rogers and Sharp (2002): “Designing interactive products to support people in their everyday and working lives.”
Jennifer Preece, Yvonne Rogers, Helen Sharp (2002) Interaction Design, ISBN: 0471492787, <http://www.id-book.com/>, Chapter 9
 - Winograd (1997): “The design of spaces for human communication and interaction.”
Winograd, T. (1997). From computing machinery to interaction design. In Denning, P., & Metcalfe, R. (Eds.), Beyond calculation: The next fifty years of computing, 149-162. Springer-Verlag. <http://hci.stanford.edu/~winograd/acm97.html>.

Interaction Design vs. User Experience Design

Definition

- “**Interaction Design** (IxD) is the design of interactive products and services in which a designer’s focus goes beyond the item in development to **include the way users will interact with it.**” <https://www.interaction-design.org/literature/topics/interaction-design>
- **User Experience Design** describes the more holistic approach in **designing the experience a person will encounter** when interacting with a devices, system or application. It considers the user interface design in the broader context of use.

Utility, Usability, Likeability

- **Utility**
a product can be used to reach a certain goal or to perform a certain task. This is essential!
- **Usability**
relates to the question of quality and efficiency. E.g. how well does a product support the user to reach a certain goal or to perform a certain task.
- **Likeability**
this may be related to utility and usability but not necessarily. People may like a product for any other reason...

What is Usability

A definition by Jakob Nielsen

Usability 101 by Jakob Nielsen

- “Usability is a quality attribute that assesses how easy user interfaces are to use. The word ‘usability’ also refers to methods for improving ease-of-use during the design process.”

The screenshot shows the NN/g Nielsen Norman Group website. The header includes the logo and tagline 'World Leaders in Research-Based User Experience'. A navigation bar lists 'Home', 'Articles' (highlighted), 'Training & Events', 'Consulting', 'Reports & Books', and 'About NN/g'. On the left, a 'Topics' sidebar lists various categories like Agile, Design Process, Ecommerce, etc. The main content area features the article title 'Usability 101: Introduction to Usability', a summary paragraph, the author 'By Jakob Nielsen on January 3, 2012', and a list of 5 quality components: Learnability, Efficiency, Memorability, Errors, and Satisfaction.

NN/g Nielsen Norman Group
World Leaders in Research-Based User Experience

Home Articles Training & Events Consulting Reports & Books About NN/g

Topics
Agile
Design Process
Ecommerce
Intranets
Navigation
Psychology and UX
Research Methods
User Testing
Web Usability
Writing for the Web
▶ See all topics

Popular Articles
10 Usability Heuristics for User Interface Design
When to Use Which User-Experience Research Methods
Empathy Mapping: The First Step in Design Thinking
Service Blueprints: Definition

Usability 101: Introduction to Usability

Summary: How to define usability? How, when, and where to improve it? Why should you care? Overview defines key usability concepts and answers basic questions.

By Jakob Nielsen on January 3, 2012 Topics: Human Computer Interaction, User Testing, Web Usability

This is the article to give to your boss or anyone else who doesn't have much time, but needs to know the basic usability facts.

What — Definition of Usability

Usability is a **quality attribute** that assesses how easy user interfaces are to use. The word "usability" also refers to methods for improving ease-of-use during the design process.

Usability is defined by **5 quality components**:

- **Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency:** Once users have learned the design, how quickly can they perform tasks?
- **Memorability:** When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors:** How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

Jakob Nielsen's Alertbox, August 25, 2003: Usability 101: Introduction to Usability
<https://www.nngroup.com/articles/usability-101-introduction-to-usability/>

Usability has five quality components

Usability 101 by Jakob Nielsen

- **Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency:** Once users have learned the design, how quickly can they perform tasks?
- **Memorability:** When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors:** How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

Jakob Nielsen's Alertbox, August 25, 2003: Usability 101: Introduction to Usability
<https://www.nngroup.com/articles/usability-101-introduction-to-usability/>

Topics

Agile
Design Process
Ecommerce
Intranets
Navigation
Psychology and UX
Research Methods
User Testing
Web Usability
Writing for the Web
▶ See all topics

Popular Articles

10 Usability Heuristics for User Interface Design
When to Use Which User-Experience Research Methods
Empathy Mapping: The First Step in Design Thinking
Service Blueprints: Definition

Usability 101: Introduction to Usability

Summary: How to define usability? How, when, and where to improve it? Why should you care? Overview defines key usability concepts and answers basic questions.

By Jakob Nielsen on January 3, 2012

Topics: Human Computer Interaction, User Testing, Web Usability

This is the article to give to your boss or anyone else who doesn't have much time, but needs to know the basic usability facts.

What — Definition of Usability

Usability is a **quality attribute** that assesses how easy user interfaces are to use. The word "usability" also refers to methods for improving ease-of-use during the design process.

Usability is defined by **5 quality components**:

- **Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency:** Once users have learned the design, how quickly can they perform tasks?
- **Memorability:** When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors:** How many **errors** do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

Usability, User Experience and Human-Centred Design

Definitions according to ISO 9241-210:2019(en)

- Usability: “extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use”
- User Experience: “user’s perceptions and responses that result from the use and/or anticipated use of a system, product or service”
- Human-Centred Design: “approach to systems design and development that aims to make interactive systems more usable by focusing on the use of the system and applying human factors/ergonomics and usability knowledge and techniques”

<https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:v1:en>



Did you understand this lesson?

Can you answer these questions?

- Describe a pragmatic computer science view point of the concept human-computer interaction.
- How can you characterize an interactive system?
- Discuss: is a video editor an interactive system?
- What is a user interface? What are similar terms? Provide Examples?
- Which ISO standard is central for human-computer interaction?
- Explain the terms utility, usability, an likeability and how they relate.
- What are the 5 quality components for usability according to Jakob Nielsen?

References

- Thomas T. Hewett, Ronald Baecker, Stuart Card, Tom Carey, Jean Gasen, Marilyn Mantei, Gary Perlman, Gary Strong, and William Verplank. 1992. ACM SIGCHI Curricula for Human-Computer Interaction. Technical Report. Association for Computing Machinery, New York, NY, USA.
- Standard, ISO 9241:210 <https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:v1:en>
- <https://www.interaction-design.org/literature/article/what-is-interaction-design> by Yu Siang Teo
- Jakob Nielsen's Alertbox, August 25, 2003: Usability 101: Introduction to Usability <https://www.nngroup.com/articles/usability-101-introduction-to-usability/>
- Jakob Nielsen. 2009. Powers of 10: Time Scales in User Experience. <https://www.nngroup.com/articles/powers-of-10-time-scales-in-ux/>
- Jennifer Preece, Yvonne Rogers, Helen Sharp (2002) Interaction Design, ISBN: 0471492787, <http://www.id-book.com/>, Chapter 9
- Winograd, T. (1997). From computing machinery to interaction design. In Denning, P., & Metcalfe, R. (Eds.), Beyond calculation: The next fifty years of computing, 149-162. Springer-Verlag. <http://hci.stanford.edu/~winograd/acm97.html>.



Terminology - UX

Learning Goals

- Understand ...
 - terms related to user experience and interaction design,
 - descriptions and texts that use these terms.
- Be able to explain technical terms, e.g.
 - Interaction design dimensions according to G. Crampton Smith
 - User experience and user experience design
 - Experience economy

User Interface Design vs. Interaction Design vs. User Experience Design

Definition

- **User Interface Design** refers to the conception and design of the **user interface**, including the choice of **modalities**, selection of **interface elements** and their **placement** in the interface.
- “**Interaction Design** (IxD) is the design of interactive products and services in which a designer’s focus goes beyond the item in development to **include the way users will interact with it.**” <https://www.interaction-design.org/literature/topics/interaction-design>
- **User Experience Design** describes the more holistic approach in **designing the experience a person will encounter** when interacting with a devices, system or application. It considers the user interface design in the broader context of use.

The Languages/Dimensions of Interaction Design

Gillian Crampton Smith in “What is Interaction Design”

- **Words**: “1-D includes **words** and poetry. [...]”
- **Visual Representations**: The 2-D languages that interaction design can borrow from include **painting**, typography, diagrams, and icons. [...]”
- **Physical Objects or Space**: 3-D languages are those of **physical**, sculptural form. [...] If something has a handle, for example, we know we are meant to grab it [...]
- **Time**: The fourth dimension is **time**. The 4-D languages include sound, film, and animation.”

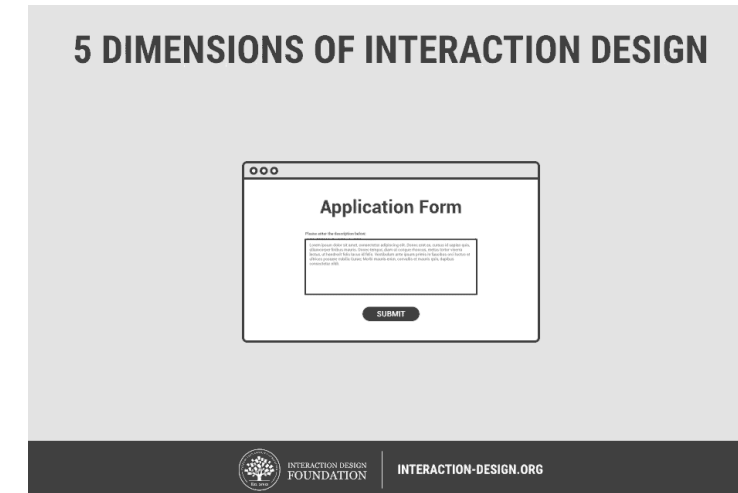
Foreword “What is Interaction Design?”. Interview with Gillian Crampton Smith. Moggridge, Bill, and Bill Atkinson. *Designing interactions*. Vol. 17. Cambridge, MA: MIT press, 2007. page xvii

The 5 Dimensions of Interaction Design

by Yu Siang Teo

<https://www.interaction-design.org/literature/article/what-is-interaction-design>

- Originally by Gillian Crampton Smith (4 dimensions); Kevin Silver added the fifth.
- 1D: Words should be meaningful and simple to understand.
- 2D: Visual representations supplement the words used to communicate information to users.
- 3D: Physical objects or space looks at what physical objects do users interact with the product (laptop, mouse, touchscreen, phones, etc.)? These all affect the interaction between the user and the product.
- 4D: Time refers to media that changes with time (e.g. animation, videos, sounds).
- 5D: **Behaviour** includes the mechanism of a product. How do users perform actions?



<https://www.interaction-design.org/>

Fashion?

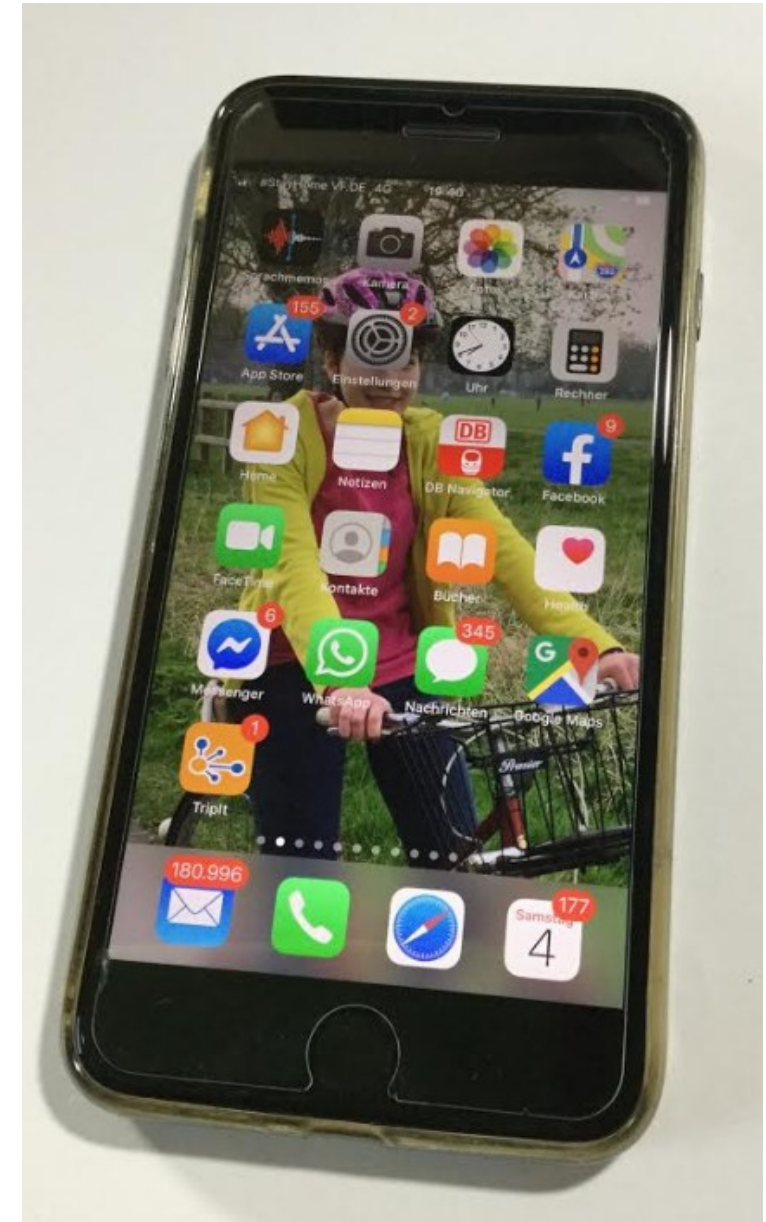
How long does an interface look new?
Is there a timeless interaction design?

IBM Simon Smartphone (1990s)



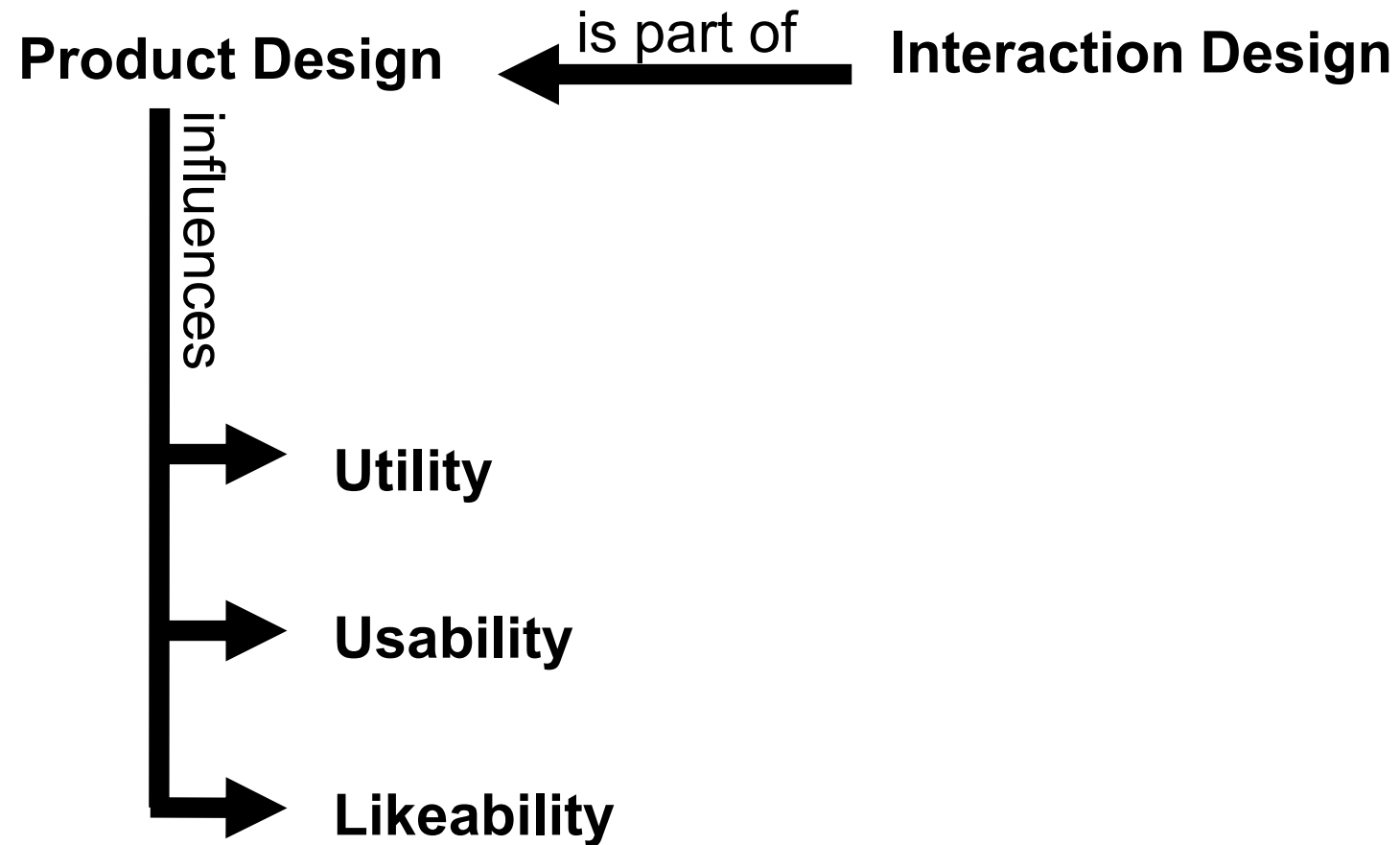
Photo by Bcos47 (public domain)
http://en.wikipedia.org/wiki/File:IBM_Simon_Personal_Communicator.png

Nokia N95 Smartphone (2007)



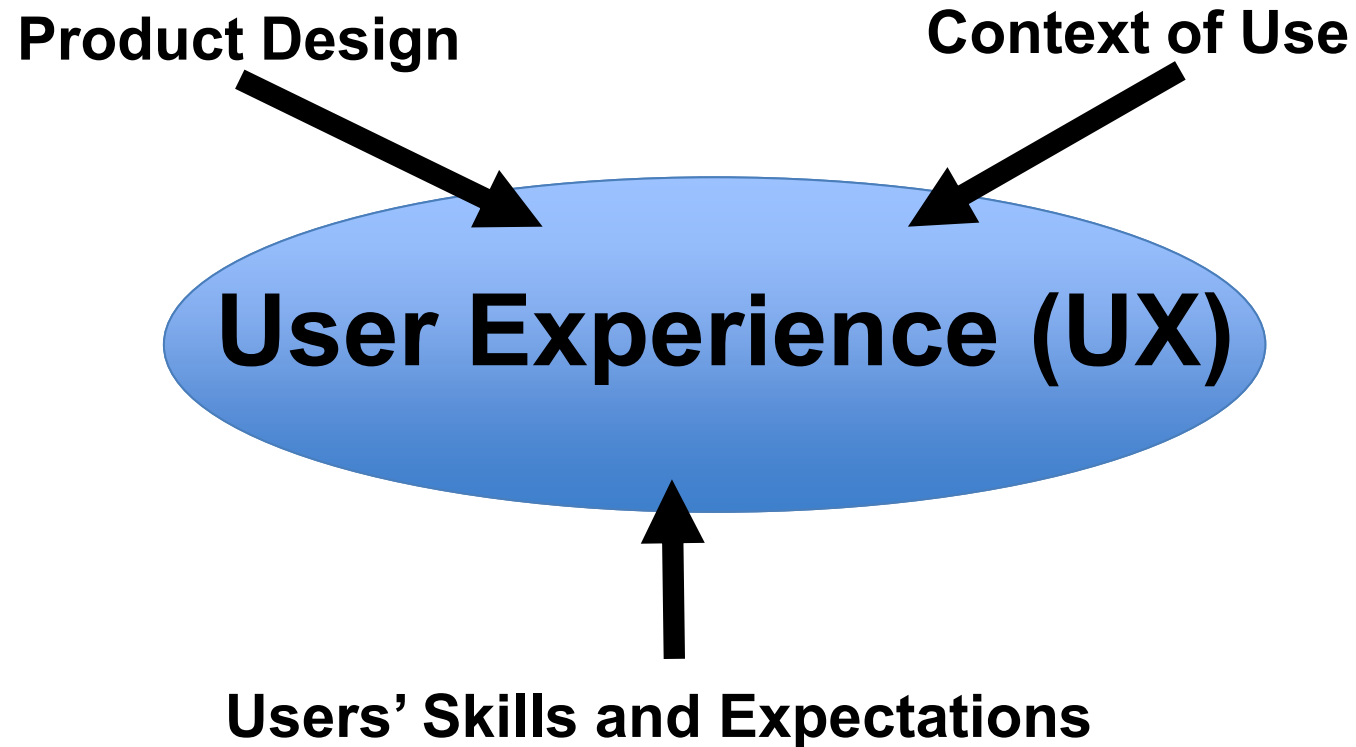
Aspects of Product Design

A simplistic view



Major Factors Influencing the User Experience

A simplistic view



One cannot not communicate

Every behavior is a form of communication.

Mini-Exercise: Find out who made this statement and in what context?

One cannot not communicate

Every behavior is a form of communication.

Mini-Exercise: Find out who made this statement and in what context?

- Paul Watzlawick
- Axioms of communication

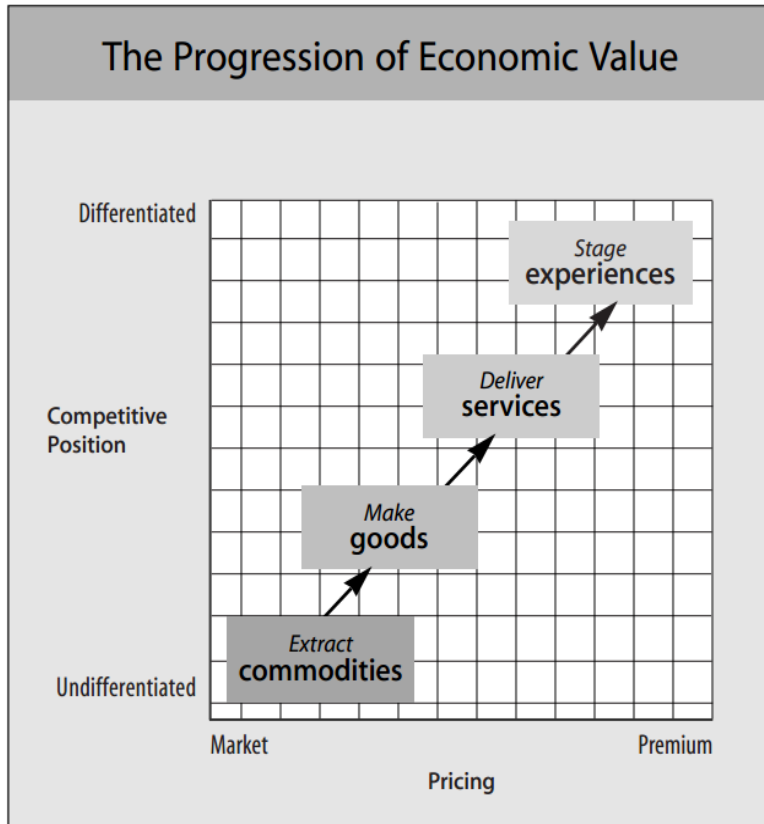
<https://www.paulwatzlawick.de/axiome.html>

You cannot not create a User Experience

Every design will create a user experience.

Experience Economy

An Economic Perspective on UX



Pine, B. Joseph, and James H. Gilmore. "Welcome to the experience economy." Harvard business review 76 (1998): 97-105. <https://hbr.org/1998/07/welcome-to-the-experience-economy>



Did you understand this lesson?

Can you answer these questions?

- Gillian Crampton Smith proposes 4 dimensions of interaction design. What are they? Provide examples of user interfaces components
- Explain the term user experience and name major factors influencing it.
- Why is user experience relevant from an economical perspective?

User Experience in Detail

Recommended Additional Material - read or watch



Video 3.1: Marc's introduction to User Experience and Experience Design.

Courtesy of Rikke Friis Dam and Mads Soegaard.
Copyright: CC-Att-ND (Creative Commons Attribution-NoDerivs 3.0 Unported). View full screen version on youtube. View transcription/captions.



Hassenzahl, Marc (2013): User Experience and Experience Design. In: Soegaard, Mads and Dam, Rikke Friis (eds.). "The Encyclopedia of Human-Computer Interaction, 2nd Ed.". Aarhus, Denmark: The Interaction Design Foundation.
http://www.interaction-design.org/encyclopedia/user_experience_and_experience_design.html

Effie Lai-Chong Law, Virpi Roto, Marc Hassenzahl, Arnold P.O.S. Vermeeren, and Joke Kort. 2009. Understanding, scoping and defining user experience: a survey approach. In Proceedings of the 27th international conference on Human factors in computing systems(CHI '09). ACM, New York, NY, USA, 719-728. DOI=10.1145/1518701.1518813
<http://doi.acm.org/10.1145/1518701.1518813>

References

- <https://www.interaction-design.org/literature/topics/interaction-design>
- Foreword "What is Interaction Design?". Interview with Gillian Crampton Smith. Moggridge, Bill, and Bill Atkinson. Designing interactions. Vol. 17. Cambridge, MA: MIT press, 2007. page xvii
- <https://www.interaction-design.org/literature/article/what-is-interaction-design> by Yu Siang Teo
- Pine, B. Joseph, and James H. Gilmore. "Welcome to the experience economy." Harvard business review 76 (1998): 97-105.
<https://hbr.org/1998/07/welcome-to-the-experience-economy>
- Hassenzahl, Marc (2013): User Experience and Experience Design. In: Soegaard, Mads and Dam, Rikke Friis (eds.). "The Encyclopedia of Human-Computer Interaction, 2nd Ed.". Aarhus, Denmark: The Interaction Design Foundation.
http://www.interaction-design.org/encyclopedia/user_experience_and_experience_design.html
- Effie Lai-Chong Law, Virpi Roto, Marc Hassenzahl, Arnold P.O.S. Vermeeren, and Joke Kort. 2009. Understanding, scoping and defining user experience: a survey approach. In Proceedings of the 27th international conference on Human factors in computing systems(CHI '09). ACM, New York, NY, USA, 719-728. DOI=10.1145/1518701.1518813
<http://doi.acm.org/10.1145/1518701.1518813>

User Needs

Learning Goals

- Understand ...
 - the complexity of user needs and the challenges to uncover them.
 - the relationship between user needs and technologies available.
 - how human needs impact the technologies we develop and how technologies influence needs
- Be able to explain ...
 - the basics of Maslow's theory of motivation
 - the concept of the task-artifact cycle and give examples

User Needs

Ask the (potential) users?

User Needs

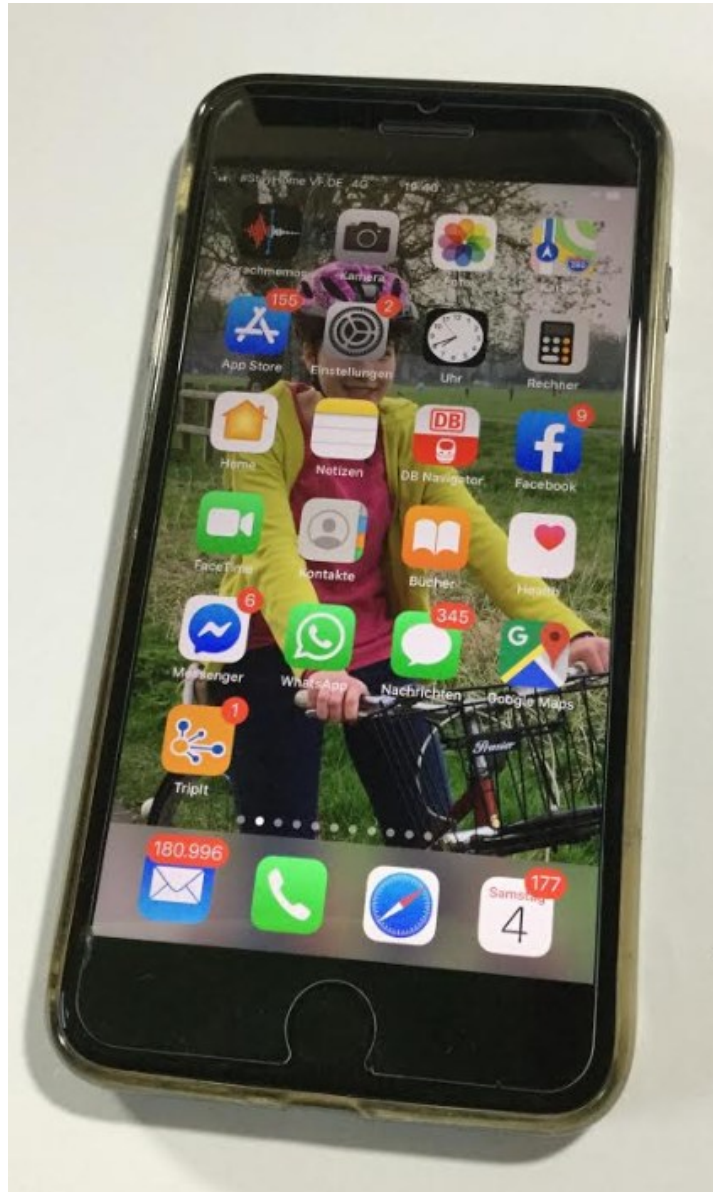
Ask the (potential) users?



User Needs

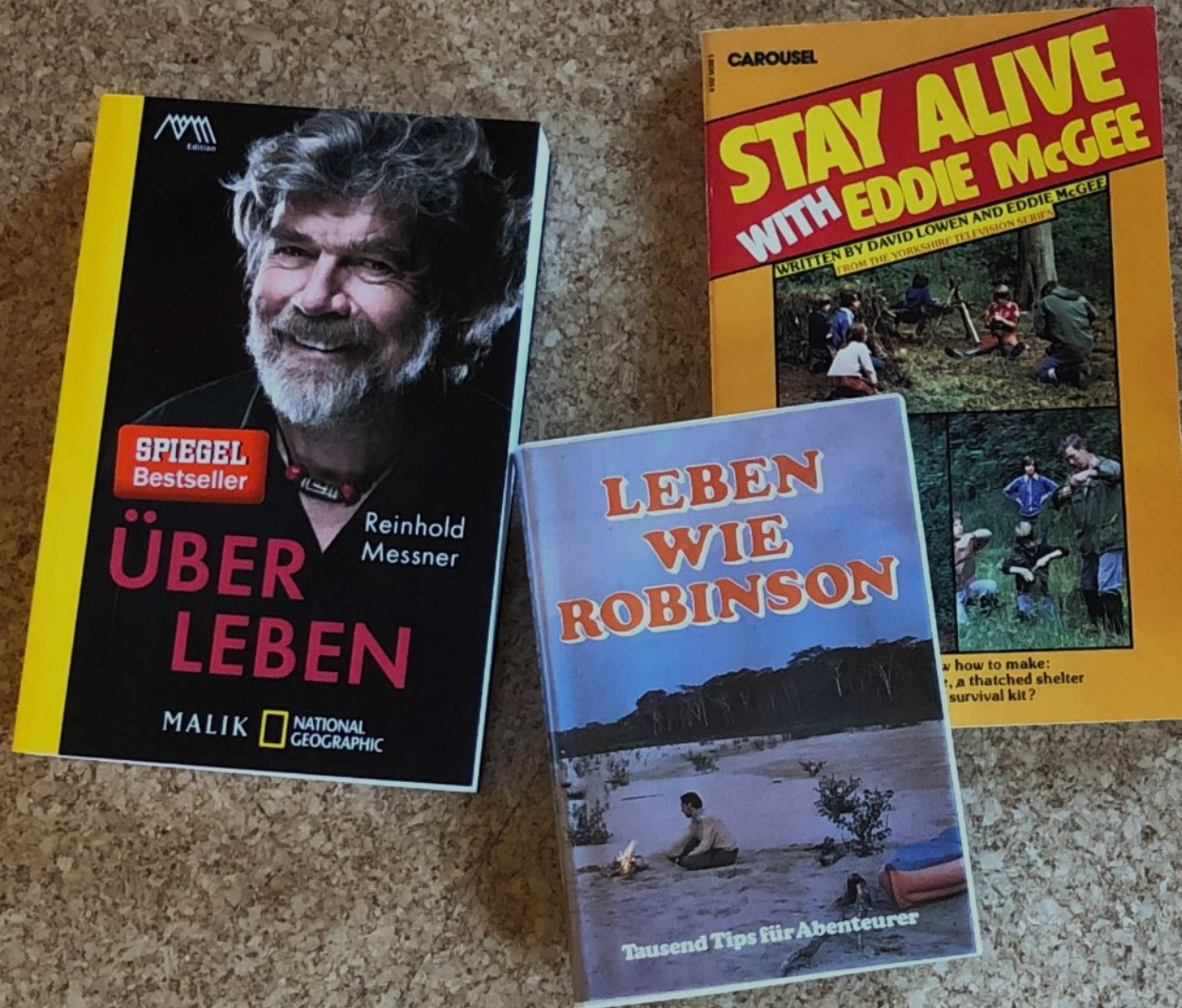
Current Technologies

Ask the (potential) users?



What do you need to survive?

- 3 Minutes
- 3 Days
- 3 Weeks



Needs

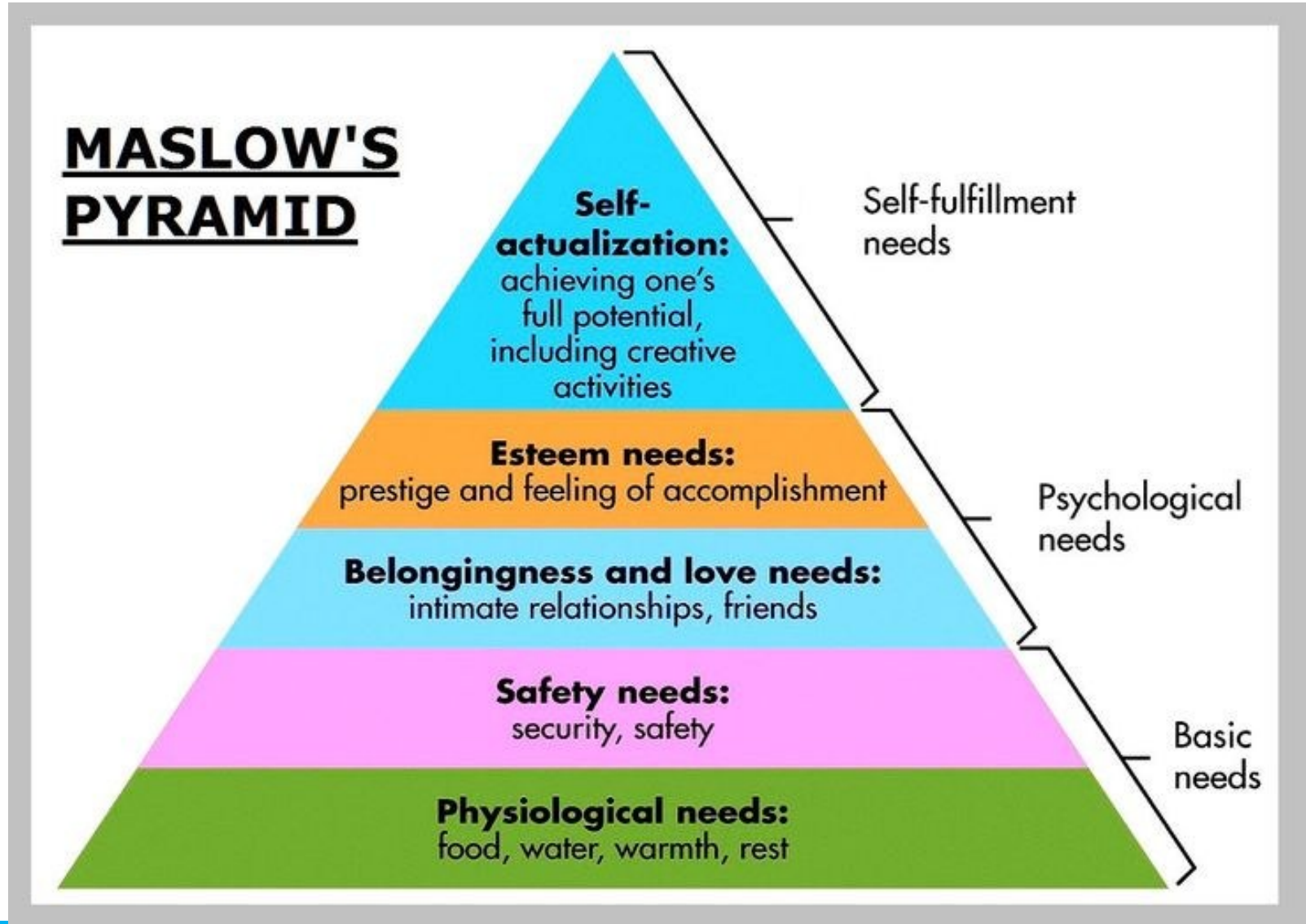
The Theory of Human Motivation

Maslow, A.H. (1943)

- 5) The Need for Self-Actualization.
- 4) The Esteem Needs
- 3) The Love Needs
- 2) The Safety Needs
- 1) The 'physiological' Needs

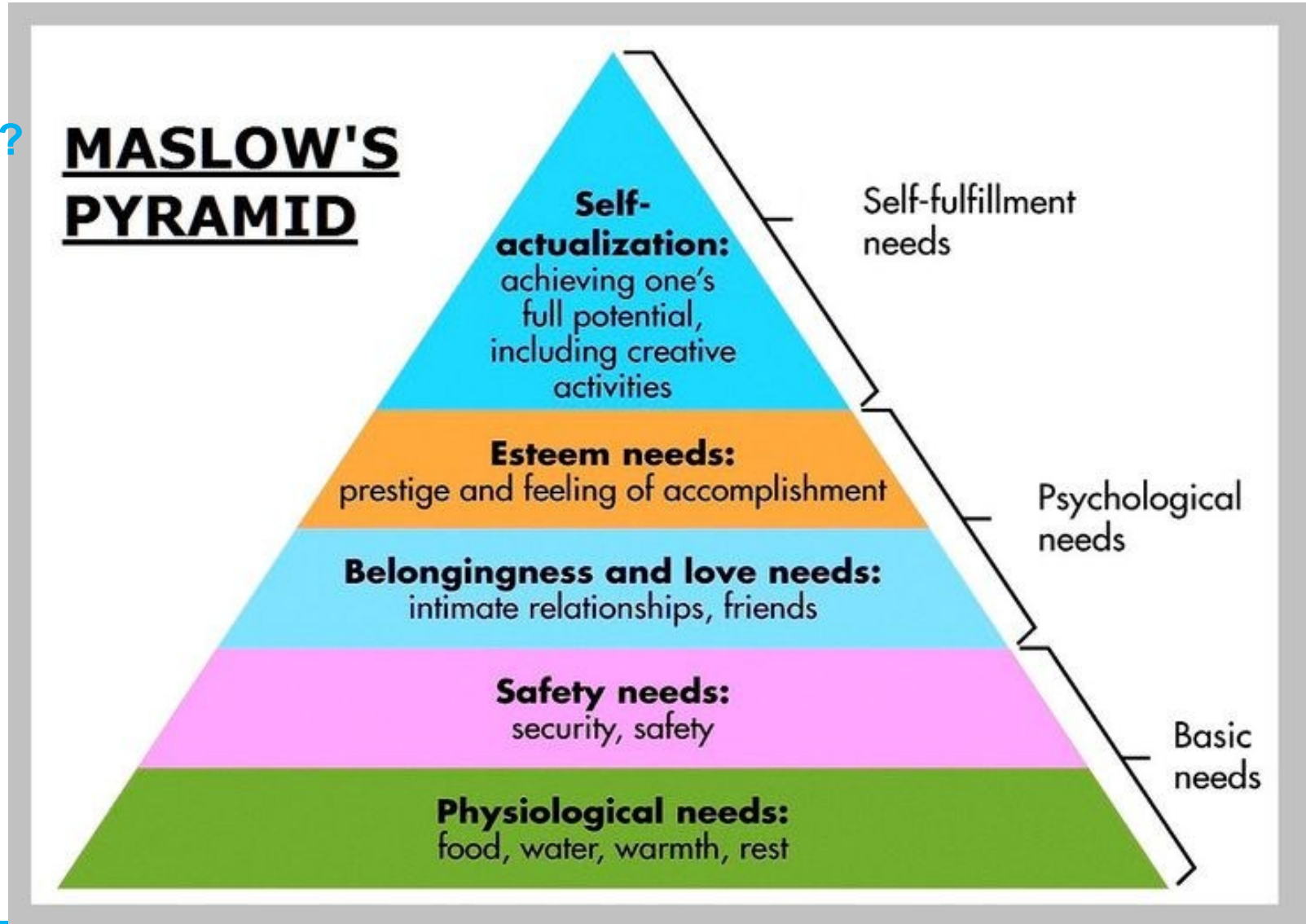
Maslow, A. H. (1943). A theory of human motivation. *Psychological Review*, 50(4), 370–396. <https://doi.org/10.1037/h0054346>

Maslow's Hierarchy of Needs



Maslow's Hierarchy of Needs

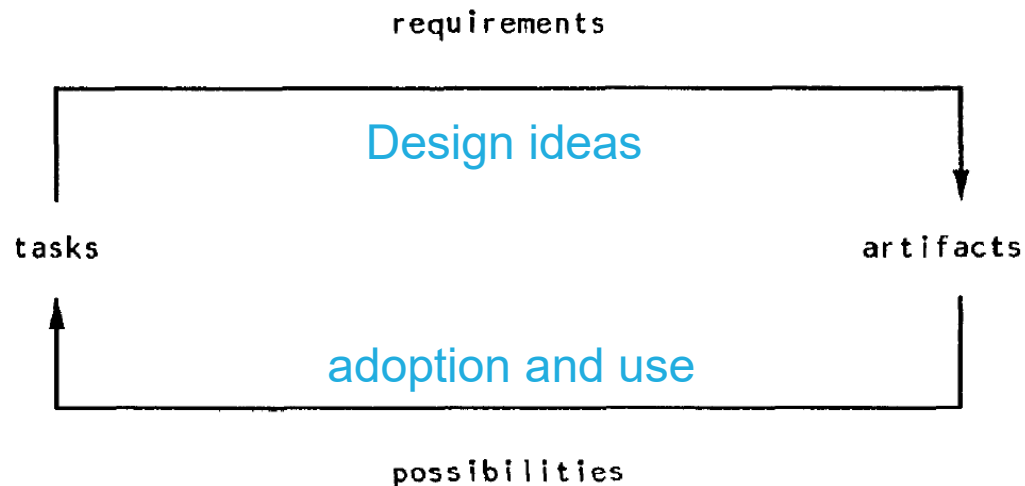
Mini Exercise: How does
Instagram, , or maps relate to this?



Task-Artifact Cycle

John Carroll 1990

- Humans have needs and preferences
- Technologies are created to suit these needs
- As humans use the technologies needs and preferences change



- **Task:** What you want to do.
- **Artifact:** The tool you use to do it.

John M. Carroll. 1990. Infinite detail and emulation in an ontologically minimized HCI. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '90). Association for Computing Machinery, New York, NY, USA, 321–328. DOI:<https://doi.org/10.1145/97243.97303>

Task-Artifact Cycle

John Carroll 1990.

“Human activities implicitly articulate needs, preferences and design visions. Artifacts are designed in response, but inevitably do more than merely respond. Through the course of their adoption and appropriation, new designs provide new possibilities for action and interaction. Ultimately, this activity articulates further human needs, preferences, and design visions.”

Carroll, John M. (2013): Human Computer Interaction - brief intro. In: Soegaard, Mads and Dam, Rikke Friis (eds.). "The Encyclopedia of Human-Computer Interaction, 2nd Ed.". Aarhus, Denmark: The Interaction Design Foundation. Available online at http://www.interaction-design.org/encyclopedia/human_computer_interaction_hci.html

Task-Artifact Cycle: Example Mobility

Need for transport → car → changed mobility behavior and town layouts



Image from OpenStreetMap:
<http://www.openstreetmap.org/?lat=48.85154&lon=10.48856&zoom=17&layers=M>

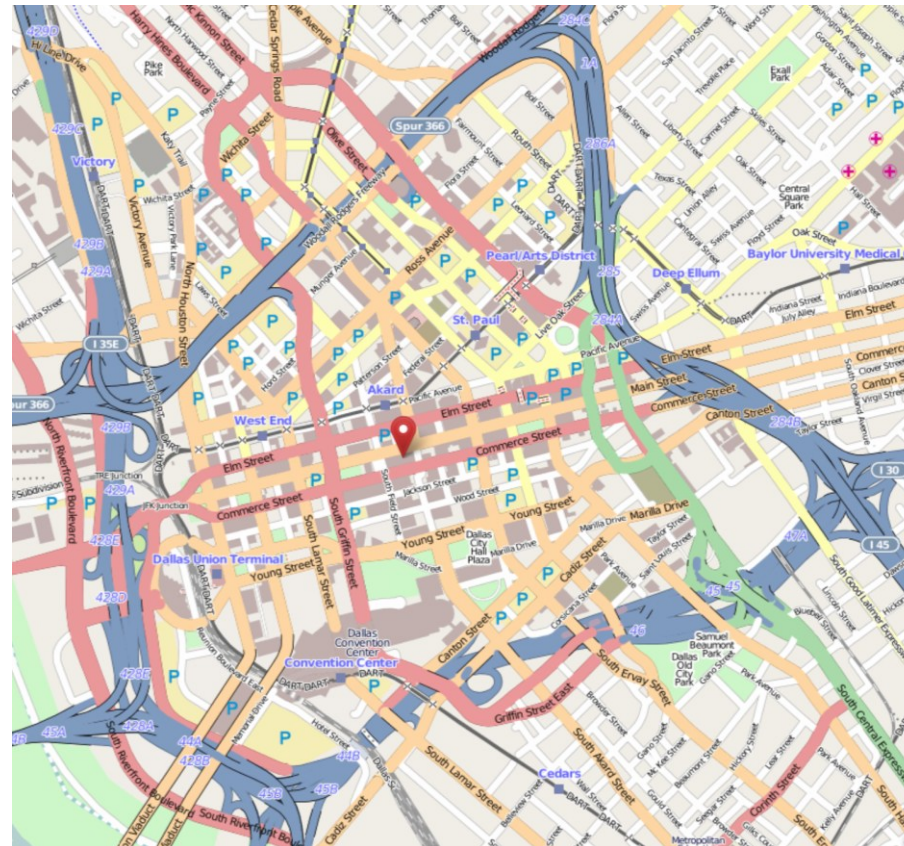


Image from OpenStreetMap:
<http://www.openstreetmap.org/?lat=49.48761&lon=8.46736&zoom=16&layers=M>

Task-Artifact Cycle

Mini Exercise: Mobile Phone

Desire to communicate

→ phone

→ changed social behavior

→ ...

1. Explain the task-artifact cycle in the context of mobile telephony.
2. How did people meet in town 1990? And how in 2024? Discuss the impact beyond a single artifact.

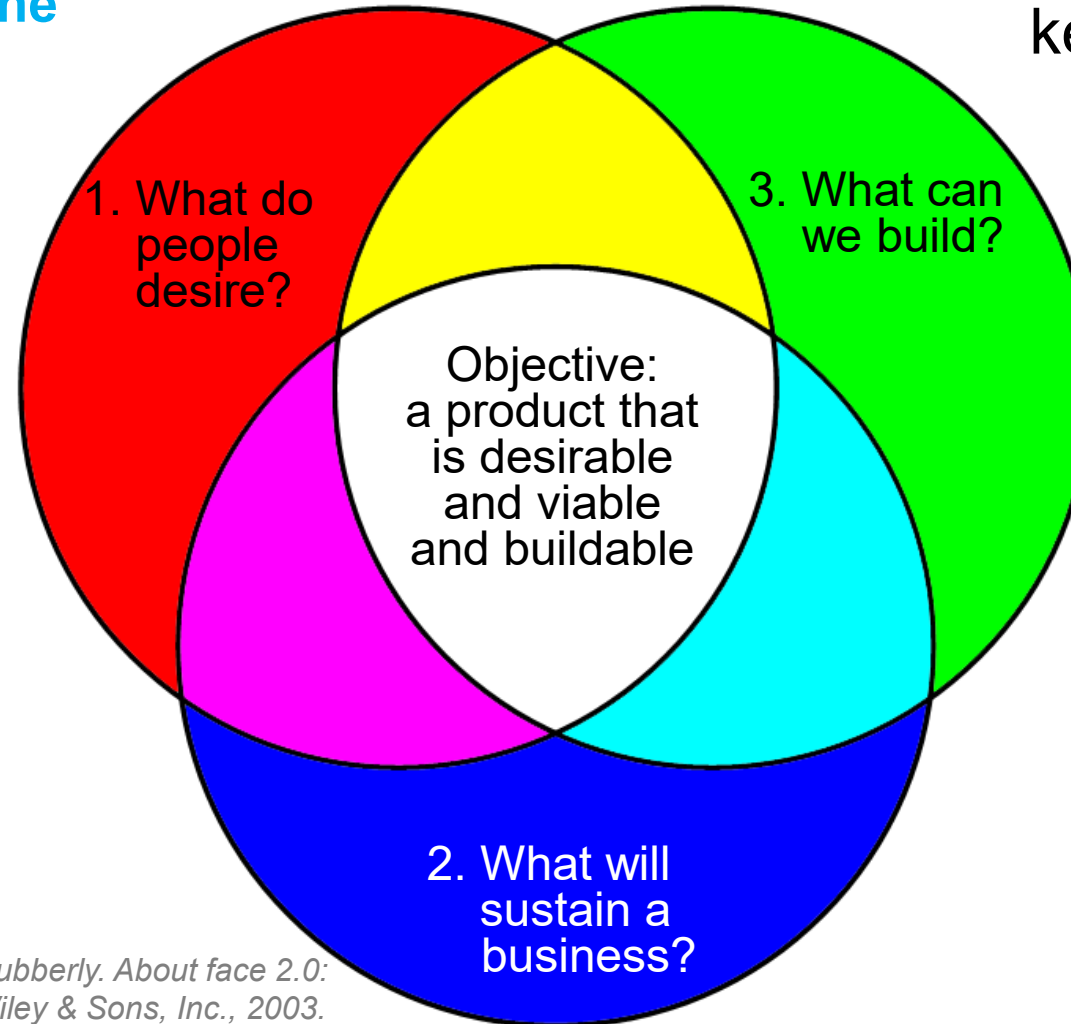


The Bigger Picture Building Successful Digital Products

It is not only about what the users want!

- Tension
 - different objectives
 - different design goals
- Context of a product design

Venn diagram illustrates the intersection of three key considerations:



Cooper, Alan, Robert Reimann, and Hugh Dubberly. About face 2.0: The essentials of interaction design. John Wiley & Sons, Inc., 2003.

Summary

- Asking the user is not sufficient to reveal user needs.
- Humans have a basic set of needs.
- Maslow states a hierarchy of needs (physiological needs, safety needs, love needs, esteem needs, Need for Self-Actualization).
- The task-artifact cycle describes how humans use the technologies changes needs.
- Understanding user needs is essential, but this is not sufficient to create a successful product.



Did you understand this block?

Can you answer these questions?

- Sketch the task-artifact cycle according to Carroll et al.
- How does this insight impact methods in human computer interaction?
- Give one real example that is explained by the task-artifact-Cycle.
- Name the levels in Maslow's hierarchy of human needs.
- Relate 3 widely use software products or web services to specific levels and explain this in one sentence each.
- How do user needs fit into Alan Cooper's concept for "Building Successful Digital Products".

References

- Maslow, A. H. (1943): A theory of human motivation. *Psychological Review*, 50(4), 370-396.
<https://doi.org/10.1037/h0054346>
- John M. Carroll. (1990): Infinite detail and emulation in an ontologically minimized HCI. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '90). Association for Computing Machinery, New York, NY, USA, 321–328. DOI:<https://doi.org/10.1145/97243.97303>
- John M. Carroll.(2013): Human Computer Interaction - brief intro. In: Soegaard, Mads and Dam, Rikke Friis (eds.). "The Encyclopedia of Human-Computer Interaction, 2nd Ed.". Aarhus, Denmark: The Interaction Design Foundation.
http://www.interaction-design.org/encyclopedia/human_computer_interaction_hci.html
- Alan Cooper, Robert Reimann, and Hugh Dubberly (2003): About face 2.0: The essentials of interaction design. John Wiley & Sons, Inc.